



Optimizing bendability of flexible electronic devices using a neutral layer strategy

Majiaqi Wu^{1,2} · Maoliang Jian² · Jianhua Zhang² · Lianqiao Yang²

Received: 2 September 2024 / Revised: 21 November 2024 / Accepted: 26 November 2024
© The Author(s) 2025

Abstract

The neutral layer (NL) strategy is a key technique for improving the bendability of flexible electronic devices. In this study, by considering a three-layer structure as an example, the results obtained by finite element analysis (FEA) showed that the NL gradually moved to the top surface of the film as the film thickness and Young's modulus increased, which are similar to the results produced by theoretical calculations. Subsequently, we optimized the thickness of a single NL structure and the failure bending radius of an indium tin oxide (ITO) electrode was reduced by 50% after optimization. In order to address the problems that affect the design of a single NL, we used optical clear adhesive (OCA) to generate multiple NLs. The FEA method was again applied to the structure and the results showed that decreasing the elastic modulus of the OCA and film thickness could reduce the maximum strain in the film. Finally, the effects of the OCA parameters on the protection of a multiple-layer ITO electrode structure were verified in bending experiments, which showed that the strain on ITO could be reduced from 5.6% to almost 0 in the two-electrode structure. The proposed strategies for designing single and multiple NLs can provide some guidance to facilitate optimizing the electronic infrastructure of flexible devices.

Keywords Flexible electronic devices · Thickness optimization design · Multiple neutral layers · Mechanical simulation

1 Introduction

Due to ongoing development and progress in science and technology, portable electronic devices such as smart phones and tablets are widely used in our daily activities and lives, and these electronic display devices are gradually tending to become more lightweight and variable in shape. The availability of a display that can be bent [1], folded [2, 3], curled [4–6], or even changed into any shape [7] allows devices to have larger screens, as well as enhancing their portability and adaptability [8], and thus many studies have investigated the production of flexible electronic display devices. An organic light-emitting diode (OLED) is a type of

dual-injection light-emitting device [9] and compared with traditional liquid crystal displays [10, 11], flexible OLEDs are characterized by their low weight and high flexibility [12–15], although technical problems still affect their reliability and durability.

Flexible OLED devices are generally produced by stacking multiple layers with specific functions, where the semiconductor functional layer plays an important role in flexible display devices [16–19]. When subjected to external loads, such as stretching, bending, and compression, the inorganic layer in the device is prone to cracking [20–23], delamination [24, 25], and buckling [23, 26, 27]. This failure caused by an external load severely detrimentally affects the electrical conductivity and structural integrity of flexible OLED devices to greatly hinder their large-scale application. Therefore, identifying a design that can effectively prevent the structural failure of a device under an applied load is crucial for ensuring the safe and reliable operation of flexible OLED devices.

Thin film encapsulation [8] is an important technique where inorganic films, such as silicon oxide (SiO_x), silicon nitride (SiN_x) film [28], or an organic/inorganic film combined composite barrier layer [29], can effectively block the

✉ Jianhua Zhang
jhzhzhang@oa.shu.edu.cn

✉ Lianqiao Yang
yanglianqiao@i.shu.edu.cn

¹ School of Microelectronics, Shanghai University, Shanghai 201800, China

² Key Laboratory of Advanced Display and System Applications, Ministry of Education, Shanghai University, Shanghai 200072, China

infiltration of water and oxygen from the environment. However, the inorganic film readily cracks when bending due to its inherent brittleness. In addition, transparent conductive films, such as indium tin oxide (ITO) films [30, 31], provide the necessary high-quality transmission and good electrical conductivity required for flexible electronic products, but they are usually brittle, and failure of the ITO film can lead to equipment failure, and affect normal operation of the device in severe cases. In addition to the buffer layer design that inhibits micro-crack propagation [32, 33], the neutral axis (NA) engineering [34] strategy can be adopted in order to protect brittle materials in devices. If the important film is placed in the neutral layer (NL), deformation of the structure can be greatly reduced, thereby effectively protecting the device.

Many studies have demonstrated that single NL position control can be achieved by adjusting the parameters of the film material in a device. For example, using a substrate with a small elastic modulus can make the NL closer to ITO [35], while the thickness of the planarized organic coating may be adjusted to make the upper SiN_x film closer to the OLED layer [36], and the thickness and elastic modulus of the cover plate can be adjusted to reduce the critical bending radius [37]. The single NL can effectively protect the single layer in the device, which is a simple design structure that also reduces the costs of calculations and usage, so this design is applied widely. However, several brittle layers in the device often need to be protected, and a single NL cannot meet the requirements described above; consequently, the multiple NLs strategy was developed. To address the multiple NLs problem, a device can be artificially divided into multiple single NL devices [38], where each has an independent NL structure, and they are then superimposed together to form an actual device with multiple NLs [39]. The introduction of multiple NLs can release the stress in each part of the device to a certain extent [40], which is generally achieved by introducing multiple adhesive layers with a low elastic modulus [41–43]. Thus, multiple NLs and single NL strategies are equally important.

In the present study, we first used the finite element analysis (FEA) method to parameterize the geometric structure and material properties of each film, before analyzing the effects of these parameters on the position of a single NL and the overall structure. Multiple simulations were conducted to obtain the optimal SiN_x film thickness for the structure containing the SiN_x encapsulation layer and ITO conductive layer. The practicability and reliability of the FEA method were verified by measuring the change in the resistance of ITO during bending, and the optimal design method for the NL thickness was obtained. Second, an optical clear adhesive (OCA) with a low elastic modulus was used to form a structure containing multiple NLs, and the FEA method was applied to explore the effects

of the OCA, encapsulated film materials, and geometric parameters on the location and number of NLs. In addition, the effect of interfacial shear was studied. Finally, the changes in the resistance of electrode structures containing one or two ITO layers during bending were tested, which verified the existence of multiple NLs in the structure that effectively protected multiple vulnerable locations.

2 Description of the model

2.1 Mechanical model of multiple films

When a flexible OLED device is bent, one or more layers with constant length before and after bending must appear in the device, which are called the strain NLs. The intersection line between the NL and cross section is the NA [44]. At the NL, the strain of the device is 0 when bent.

Suo et al. [45] provided the strain distribution expression for double-layer stacked films, and Lee et al. [37] gave the strain distribution expression for three-layer stacked films based on this expression. According to their structure, we modeled a three-layer stacked film (see Fig. 1a) and built a rectangular coordinate system with the bottom layer as the base. The thicknesses of the three-layer structure are denoted as h_s , h_c , and h_f from the bottom to the top, respectively, and the subscript s, c, and f represent the substrate, conductive layer, and film. The top strain of the film farthest from the bending center is ϵ_{top} , and the bottom strain of the substrate closest to the bending center is ϵ_{bot} . The two bending modes are also shown in Fig. 1. The tensile state of the film is convex-type bend (see Fig. 1b), and concave-type bend when the film is compressed (see Fig. 1c). Under concave-type bend, the bottom layer is defined on the film instead of the substrate.

In general, the dimension of the multilayer film composite structure is much larger in the plane direction than the thickness direction, so the cylindrical bending condition can be used, which is similar to the plane stress condition applied in the thickness direction of the structure [35]. In order to better establish the corresponding mathematical model, it is necessary to make some assumptions. First, it is assumed that each layer of materials has no defects and good adhesion between interfaces, and the materials in each layer are isotropic materials. Second, we assume that the bending radius is much larger than the thickness of the stacked layer. Finally, it is assumed that the film structure is in a state of pure bending, so the bending strain exhibits a linear distribution along the thickness direction. Thus, according to Lee et al. [37], the top and bottom strain can be expressed as:

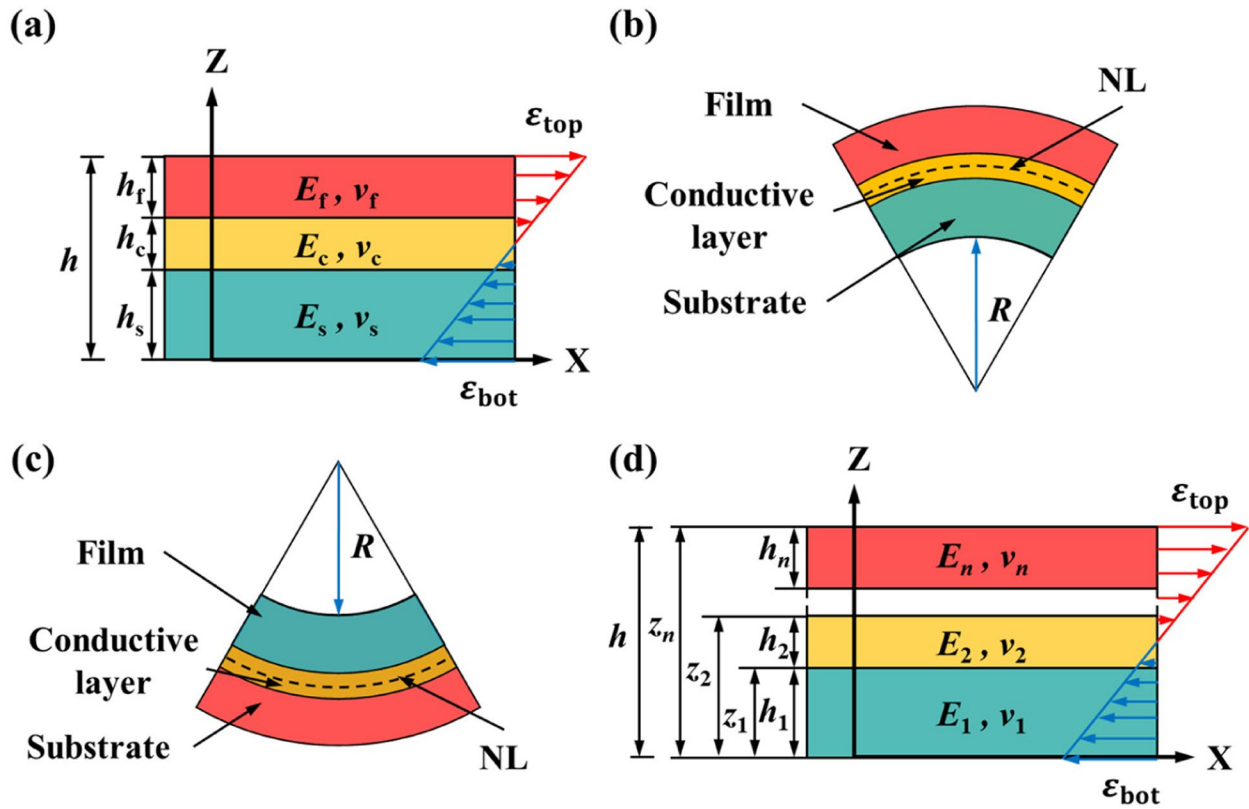


Fig. 1 **a, d** Flexural strain distributions for three-layer and multilayer stacked films and assumed entire thickness, respectively. **b, c** Stress states of three stacked films under convex-type and concave-type bend, respectively

$$\epsilon_{top} = \frac{h_s^2 \bar{E}_s + h_c^2 \bar{E}_c + h_f^2 \bar{E}_f + 2h_s h_c \bar{E}_s + 2h_c h_f \bar{E}_c + 2h_s h_f \bar{E}_s}{2R(h_s \bar{E}_s + h_c \bar{E}_c + h_f \bar{E}_f)} \quad (1)$$

$$\epsilon_{bot} = -\frac{h_s^2 \bar{E}_s + h_c^2 \bar{E}_c + h_f^2 \bar{E}_f + 2h_s h_c \bar{E}_c + 2h_c h_f \bar{E}_f + 2h_s h_f \bar{E}_f}{2R(h_s \bar{E}_s + h_c \bar{E}_c + h_f \bar{E}_f)} \quad (2)$$

where $\bar{E}_s = E_s / (1 - \nu_s^2)$, $\bar{E}_c = E_c / (1 - \nu_c^2)$, $\bar{E}_f = E_f / (1 - \nu_f^2)$, E and ν represent the Young's modulus and Poisson's ratio for the material, respectively, and R is the bending radius at the bottom. Now, we consider the structure with more than three layers in the actual device, and the multi-layer structure also satisfies the assumptions given above. For the composite structure with a total of n layers, assume that the thickness of each layer is h_i ($i = 0, 1, 2, \dots, n$), the total thickness is h , the height from the bottom layer to the top of layer i is z_i , and the bending radius at the bottom is R (see Fig. 1d). According to the force balance conditions, the expression for the bottom strain can be obtained as follows [35].

$$\epsilon_{bot} = -\frac{1}{2R} \left[\frac{\sum_{i=1}^n \frac{E_i}{1-\nu_i^2} (z_i^2 - z_{i-1}^2)}{\sum_{i=1}^n \frac{E_i}{1-\nu_i^2} (z_i - z_{i-1})} \right] \quad (3)$$

2.2 Constitutive model of OCA

In order to distribute multiple NLs in flexible OLED devices, OCA is generally used to bond each layer in the structure. OCA has high bonding properties [46] as well as strong weathering resistance and noncorrosive properties [47]. In addition, OCA is a hyperelastic and viscoelastic material, which is widely used in flexible electronic devices due to its excellent properties.

OCA is hyperelastic and a variety of material models, such as the Neo-Hookean [48], Mooney-Rivlin [49], and Yeoh [50] models, can be used to describe the hyperelastic properties of materials. Each model defines the strain energy density function U in different ways. Compared with the first two models, the Yeoh model is better at describing the

stiffening effect [51] of materials, and the fitted strain range is larger. Therefore, the Yeoh model was used in this study to describe the hyperelasticity of OCA. The model can be expressed as [3]:

$$U = \sum_{i=1}^3 C_{i0} (I_1 - 3)^i + \sum_{i=1}^3 \frac{1}{D_i} (J_{el} - 1)^{2i} \quad (4)$$

where C_{i0} and D_i are material constants, I_1 is the first partial strain invariant, and J_{el} is the volume ratio. For a completely incompressible material, $D_i = 0$, $J_{el} = 1$, and the initial shear modulus of the material is $G = 2C_{10}$.

Assuming that the OCA used is isotropic and completely incompressible, the simple shear stress σ_s can be expressed as the partial derivative of U with respect to the simple shear strain γ_s during simple shear deformation:

$$\sigma_s = \frac{\partial U}{\partial \gamma_s} = 2 \sum_{i=1}^3 i C_{i0} \gamma_s^{2i-1} \quad (5)$$

The viscoelastic behavior of materials can be described well by the generalized Maxwell model [52]. Therefore, both the time-dependent shear modulus $G(t)$ and bulk modulus $K(t)$ can be expanded into Prony series [53]:

$$\frac{G(t)}{G_0} = g(t) = 1 - \sum_{i=1}^n g_i \left[1 - \exp\left(-\frac{t}{\tau_i^G}\right) \right] \quad (6)$$

$$\frac{K(t)}{K_0} = k(t) = 1 - \sum_{i=1}^n k_i \left[1 - \exp\left(-\frac{t}{\tau_i^K}\right) \right] \quad (7)$$

where G_0 and K_0 are the instantaneous shear modulus and bulk modulus, respectively, determined by the instantaneous elastic modulus E_0 ; τ_i is the relaxation time constant; g_i and k_i are the Prony constants; and $g_i = G_i/G_0$, $k_i = K_i/K_0$.

OCA is also viscoelastic, so Prony series can be used to describe its stress relaxation process. The equilibrium Poisson's ratio for the OCA is 0.499, which is very close to 0.5, and thus it can hardly be compressed, and $\tau_i^G = \tau_i^K = \tau_i$ can be assumed at this time. The stress relaxation behavior is dominated by shear stress relaxation, so it is only necessary to conduct shear stress relaxation experiments. In general, the experimental data can be fitted better by intercepting 3–5 Prony series.

2.3 FEA simulations

The FEA simulations were conducted using ABAQUS 6.14. The bending process of the flexible device was regarded as a process where the device rotated around a central axis, and the two ends of the device were supported by a rigid plate and rotated with it, forming a 90° bending state after

the rigid plate rotated 90° at both ends, and the distance between the rigid plates was the bending diameter of the flexible device (see Fig. 2a). Subsequently, the bending radius of the flexible device was controlled by manipulating the inward movement of the rigid plate (see Fig. 2c). When the bottom strain is calculated by using Eq. (3), R is equal to the difference between the spacing of the semi-rigid plates and the total thickness of the device.

The simulation model was established according to the actual bending process for the foldable flexible device. It was assumed that the total length of the device was 60 mm and it was modeled as a symmetric device along the axis. When bending and limiting the degree of freedom along the x -axis of symmetry, the rigid plate was modeled as a rigid body and it rotated 90° clockwise around the reference point. The reference point was on the contact surface between the rigid plate and flexible device, and the distance from the axis of symmetry was the bending radius (see Fig. 2b). The gradual radius reduction process (see Fig. 2d) was obtained by restart analysis. A tetrahedral mesh with a size of 0.04 mm was used, and each film was divided into three layers. The grid element type adopted plane strain, hybridization, and reduced integration to avoid shear locking.

3 Experimental methods

The ITO layer in flexible OLED devices generally acts as a transparent conductive electrode, which is damaged when subjected to stretching and bending, and the conductivity of ITO is reduced. ITO is generally located at the top or bottom of the device's structure and its Young's modulus is large, but the thickness is usually only tens of nanometers, so the factors described above make the ITO layer more prone to failure than other structural layers. When the packaging layer, ITO layer, and substrate of the device are extracted separately to form a flexible electrode with a simple structure, the conductivity and failure mode of the electrode under an external bending load can be readily studied. Therefore, the structure of a flexible ITO electrode was investigated in the present study. Polyimide (PI) was selected as the substrate, ITO as the conductive layer, and SiN_x film was used as the encapsulation layer in some structures, with OCA as the adhesive layer in order to obtain a stable flexible electrode structure.

3.1 Preparation of flexible electrode

The ITO and SiN_x /ITO composite films on self-supported PI were prepared as shown in Fig. S1 (available in the online supplementary data). The self-supported PI adhered to the glass substrate was placed in a magnetron sputtering instrument, and the target atoms were sputtered onto the surface

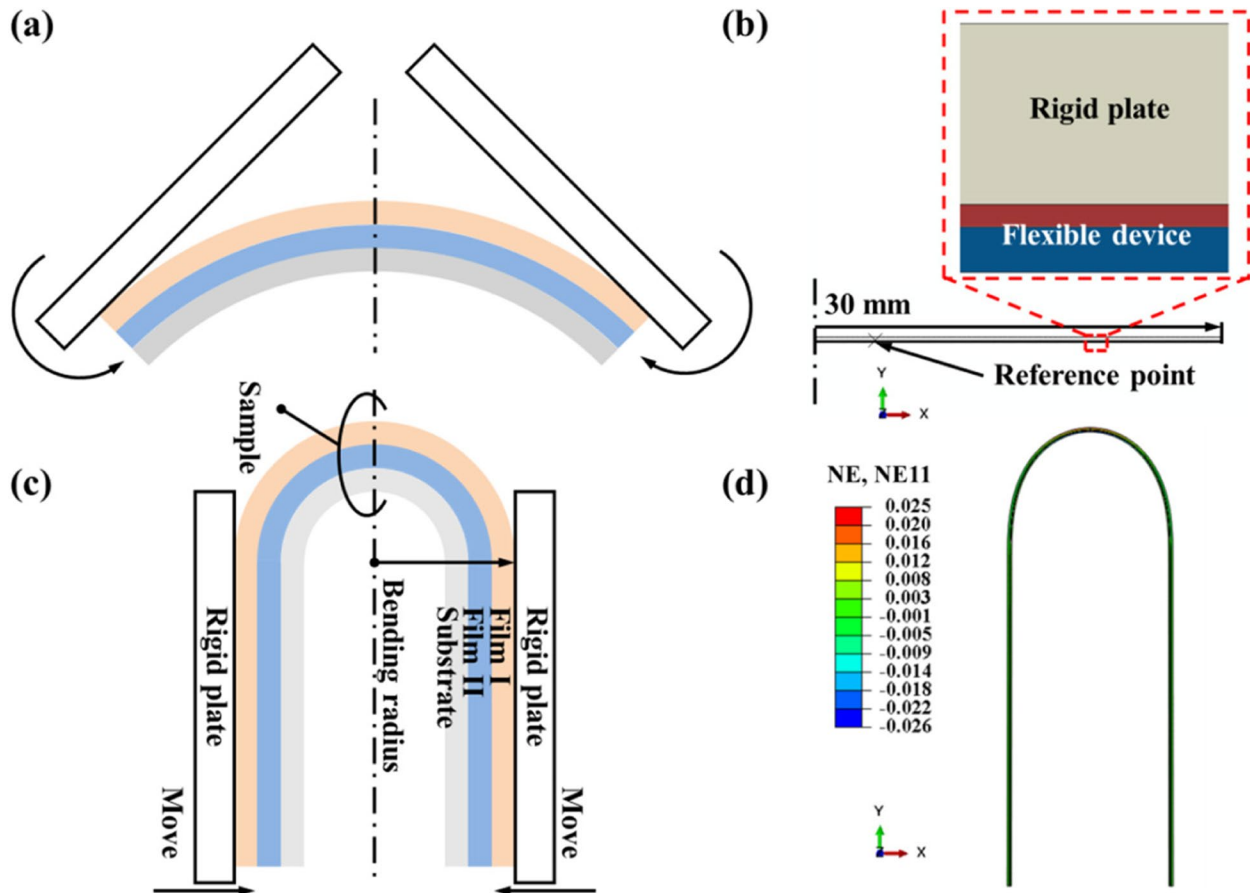


Fig. 2 Actual bending and simulation process for flexible devices. **a, b** The device rotates with the rigid plate in real situations and simulations. **c, d** Devices in real situations and simulations (rigid plate hidden) move inward with the rigid plate

of the PI substrate through Ar ion bombardment to form ITO/PI composite films (see Figs. S1a, b). The PI substrate and ITO film were cut together into a rectangle (10 mm width $\times$ 60 mm length), and the PI substrate was stripped from the glass substrate to form a single-layer electrode structure (see Fig. S1c). In addition, a layer of SiN_x film was grown on the surface of ITO (see Fig. S1d) by cutting and stripping to prepare an SiN_x /ITO/PI multilayer composite film (see Fig. S1e).

3.2 Mechanical properties of film and substrate

SiN_x films were deposited on the substrate via plasma enhanced chemical vapor deposition (PECVD; Japan, ULVAC, CME-200E). The Young's modulus was obtained for the films (~ 249 GPa) using a nanoindenter (USA, HYSTRON, TI 950 TriboIndenter) at a maximum load of 1.5 mN with a contact depth of ~ 60 nm. The Poisson's ratio could not be obtained directly for the SiN_x films, so a value of 0.20 was applied, as described in a previous study [54]. The ITO film was prepared by using the magnetron

sputtering method (Japan, ULVAC, SME-200E), and the thickness of the prepared film was 30 nm. The Young's modulus was ~ 120 GPa [55] for the ITO film and Poisson's ratio was 0.30.

The thicknesses were measured for four PI substrates (11 μm , 25 μm , 50 μm , and 100 μm) by using a step meter (Germany, BRUKER, DEKTAK XT), and the stresses were determined for the PI substrates (10 mm width $\times$ 50 mm length) under different strains with an in situ tensile testing machine (USA, AMETEK, LS5). The Young's moduli were obtained as 5.00 GPa, 1.49 GPa, 1.44 GPa, and 1.21 GPa for the four PI substrates with thicknesses of 11 μm , 25 μm , 50 μm , and 100 μm , respectively. The Poisson's ratio was obtained for the substrate by using a video extensometer (UK, LLOYD, VE-7100), and the average value was about 0.29.

3.3 Mechanical properties of OCA

The hyperelasticity and viscoelasticity were measured for the OCA (Mitsubishi) by using a rotating rheometer

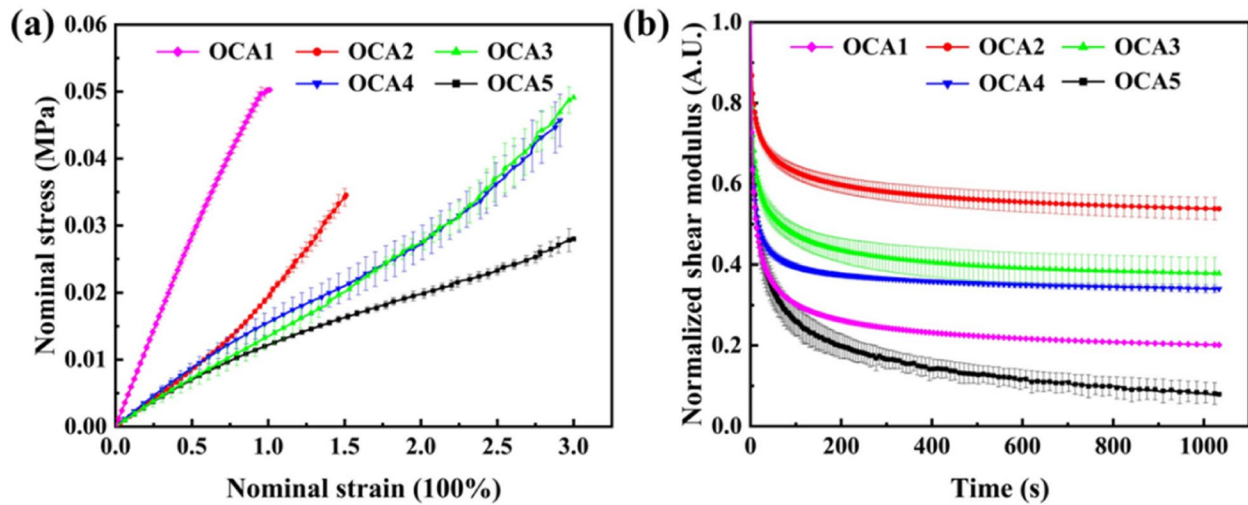


Fig. 3 a, b Nominal stress–strain and shear stress relaxation curves for OCA, respectively

Table 1 Hyperelasticity and thickness parameters for five types of OCA

OCA model	C_{10} (MPa)	C_{20} (MPa)	C_{30} (MPa)	Thickness (μm)
OCA1	2.99×10^{-2}	-1.71×10^{-3}	-3.12×10^{-4}	300
OCA2	7.66×10^{-3}	1.30×10^{-3}	-1.13×10^{-4}	125
OCA3	5.82×10^{-3}	1.85×10^{-4}	-2.81×10^{-6}	125
OCA4	8.18×10^{-3}	-2.99×10^{-4}	2.67×10^{-5}	125
OCA5	6.69×10^{-3}	-3.51×10^{-4}	-3.51×10^{-4}	25

(Austria, ANTON PAAR, MCR 302e). Each sample was measured three times and the response curve was plotted for the nominal stress and strain (see Fig. 3a). Figure 3a shows the nonlinear elastic behavior of the OCA during deformation, where the OCA appeared to exhibit stiffening behavior (the curve shifted upward) at large strains. The viscoelasticity of OCA was also measured by using a rotating rheometer. Each sample was kept under a small strain (10%) and the stress relaxation experiment was conducted in the shear mode (see Fig. 3b). The test was performed three times and the mean value was taken as the response of the modulus and time. The stress relaxation process entails stress decaying with time under constant strain. In general, stress decays exponentially as time increases and this behavior can

be characterized in stress relaxation experiments. Figure 3b also demonstrates that the stress does not relax to 0 for viscoelastic solids, such as OCA.

According to the Yeoh model of simple shear stress–strain, three material parameters were obtained for the five types of OCA by fitting the mean values in the shear stress–strain curves, as shown in Table 1. Material parameters C_{20} and C_{30} had negative values, and thus the fitted stress–strain curve was S-shaped and it captured the strain stiffening effect of the material [51]. Table 1 also shows that the initial shear modulus was largest for OCA1. Among the three types of OCA with thicknesses of 125 μm , the initial shear moduli were similar for OCA2 and OCA4, but the former had a larger shear modulus at higher strains. The initial shear modulus was lower for OCA3 than OCA4, but both were very similar at higher strains.

According to the generalized Maxwell model, after importing a series of experimental data into ABAQUS, the program automatically fitted the curve to yield the third-order Prony and relaxation time constants g_i and τ_i^G , respectively, as shown in Table 2.

3.4 Bending tests

During the bending process, ITO films may be subjected to tensile or compressive stress. The accurate control of the

Table 2 Viscoelastic parameters for five types of OCA

i	OCA1		OCA2		OCA3		OCA4		OCA5	
	g_i	τ_i^G	g_i	τ_i^G	g_i	τ_i^G	g_i	τ_i^G	g_i	τ_i^G
1	0.450	4.23	0.214	4.22	0.370	3.99	0.470	3.87	0.439	5.18
2	0.220	33.5	0.131	35.4	0.152	32.7	0.135	31.0	0.258	36.2
3	0.130	275	0.115	331	0.102	279	0.052	412	0.225	294

bending radius and measuring its conductive properties in a stable manner were then investigated in this study. The bending test was carried out using a dynamic bending test device, which was manufactured in the laboratory, as shown in Fig. S2.

4 Results and discussion

4.1 Effect of packaging film on NL

For the three-layer structure shown in Fig. S1e, SiN_x films with different thicknesses and Young's moduli were obtained by changing the deposition time and process parameters, and then used as encapsulation layers in the structure. In order to

Table 3 Material properties and film thicknesses used in FEA for flexible encapsulation

Labels (from top to bottom)	Material	$E(\text{GPa})$	ν	$h(\mu\text{m})$
Film	SiN_x film	5	0.29	50/75/100/125/150
ITO	ITO	120	0.30	0.03
Substrate	PI substrate	5	0.29	100

explore the effect of the film thickness on the NL, the material parameters were kept unchanged for the ITO layer and substrate. In order to simplify the calculations, the Young's modulus and Poisson's ratio for the film were consistent with those for the substrate, and only the thickness of the film changed. The material parameters for the film layer, ITO layer, and substrate are listed in Table 3, and convex-type bend simulations with an initial bending radius of 11 mm were conducted for five films with different thicknesses.

After conducting the simulations, the nominal strains on the nodes at the symmetric interface of the model along the x -axis direction were extracted to obtain the strain distribution along the thickness direction of the structure (see Fig. 4a). The linear fitting method was used to fit these data and the linearity of the data was very high, which is consistent with the assumption that the strain exhibited a linear distribution along the thickness direction in the theoretical method. The simulation results obtained for the concave-type bend are shown in Fig. 4d. Compared with the results obtained for the convex-type bend, the nominal strains at the same thickness were the opposite under the two bending modes.

As shown in Fig. 4b, a simulated strain cloud map for the film was placed evenly in a bar chart representing the total thickness of the structure, as well as the position of

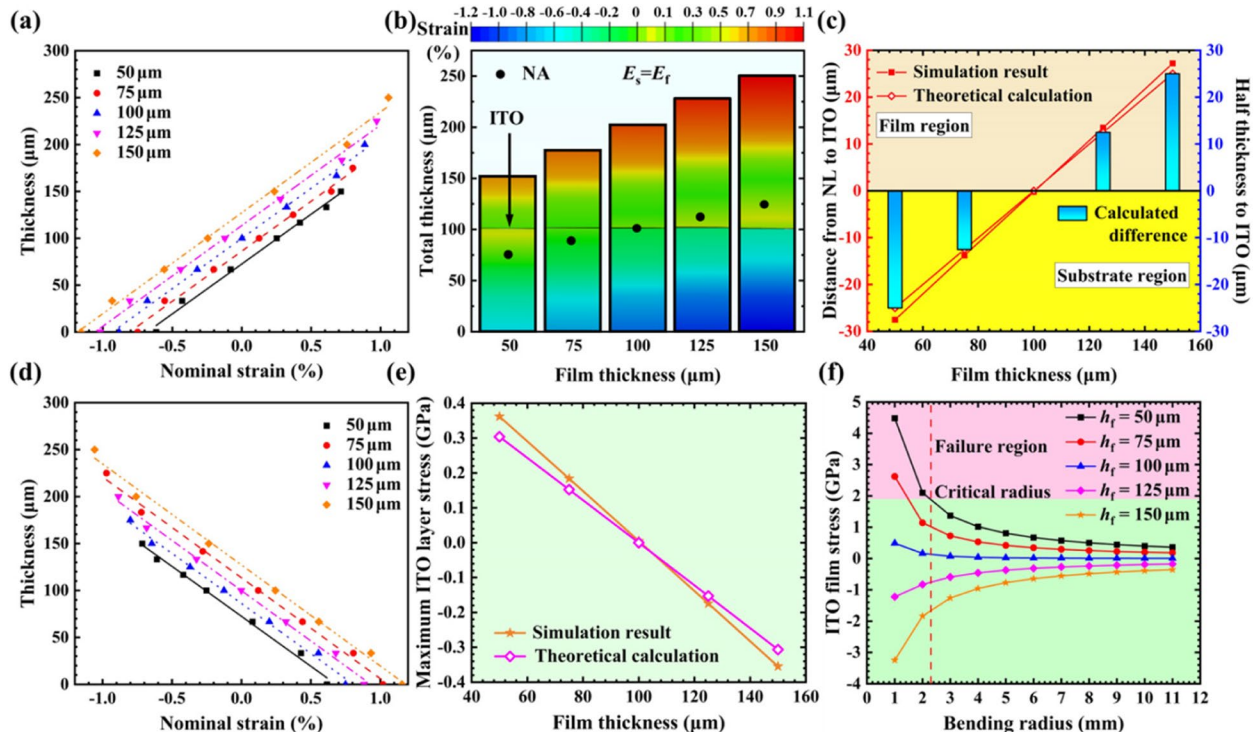


Fig. 4 Strain distribution in the thickness direction with different film thicknesses: **a** convex-type and **d** concave-type bends. Position of NL and stress distribution in structure of ITO film: **b** position of NA in

thickness direction, **c** distance from NL to the ITO film and difference between the half-thickness position and ITO film, **e** maximum stress in ITO film, and **f** ITO film stress under different bending radii

the NA. As the thickness of the SiN_x film increased, NA moved toward ITO and the tensile strain of ITO decreased gradually. NA was approximately coincident with ITO when the thickness was 100 μm , but as the thickness of the film continued to increase, ITO moved away from NA and the compressive stress increased. As shown in Fig. 4c, the distance from NL to ITO was defined as negative when NL was below ITO. As the thickness of the film increased, NL expanded from the substrate region ($-27.5 \mu\text{m}$) to the film region. When the film thickness was 100 μm , ITO was only 0.2 μm away from NL. The simulation results were highly consistent with the theoretical calculations. The bar chart in Fig. 4c shows the difference between the ITO and half-thickness position in the device. The simulation results showed that when the Young's moduli were the same for the film and the substrate, NL appeared at half of the total thickness. As the film thickness increased, the stress in the ITO layer was characterized by an extreme stress of about -0.36 GPa (see Fig. 4e). According to these results, we consider that the movement of the NL made the ITO gradually lower than the NL and compressive stress was generated.

According to the bending experiments described in Sect. 4.3, 1.9 GPa was defined as the failure stress for ITO when damaged by stretching. As shown in Fig. 4f, the stress in the ITO layer gradually increased as the bending radius decreased further. For the film with a thickness of 50 μm , when the bending radius was less than 2.2 mm, the stress in the ITO layer exceeded 1.9 GPa, and the bending radius at this time was defined as the critical radius when the stress was greater than 1.9 GPa, and ITO was in the failure region. The stress on ITO in the film with a thickness of 100 μm was smallest at all bending radii because the NL in the structure was closest to the ITO. Thus, when designing the structure, a reasonable film thickness should be selected to ensure that the ITO is still under low stress at a smaller bending radius.

To explore the effect of the Young's modulus of the film on the NL, the thickness of the film was kept unchanged (50 μm) and only the Young's modulus was changed (Table 4), whereas the elastic parameters and thicknesses of the substrate and ITO layer were kept unchanged. A bending simulation was also carried out for the structure with an initial bending radius of 11 mm.

Clearly, for films with different Young's moduli, the strain in the device also satisfied the assumption of a linear distribution in the thickness direction (see Fig. 5a). In addition, as the Young's modulus increased, the maximum compressive strain increased whereas the tensile strain decreased. As the bending radius decreased, the maximum compressive stress and tensile stress increased, but the fitted curves intersected at a point where the strain was 0, and this intersection point was the NL (see Fig. 5d), thereby indicating that the position of the NL did not change with the bending radius.

Table 4 Material properties and Young's moduli (E) of films used in FEA for flexible encapsulation

Labels (from top to bottom)	Material	$E(\text{GPa})$	ν	$h(\mu\text{m})$
Film	SiN_x film	5/10/15/20/25	0.29	50
ITO	ITO	120	0.30	0.03
Substrate	PI substrate	5	0.29	100

As the Young's modulus of the film increased, the position of the NA also gradually increased from below ITO at the beginning to near ITO at 20 GPa and higher than ITO at 25 GPa (see Fig. 5b). Only the Young's modulus of the film changed but the changes in the positions of the NL were same as those when the thickness increased, thereby demonstrating that changing the Young's modulus could also make the NA move toward the ITO. In addition, the NA was evidently not in the middle of the device but instead it became closer to the film as the Young's modulus increased. As shown in Fig. 5c, the simulation results were still highly accurate. When the Young's modulus of the film was 20 GPa, the NL was very close to the ITO and the Young's modulus at this point was the critical modulus. In the practice, the structure should be made close to the critical modulus of the design.

Figure 5e shows that the differences between the simulation results and theoretical calculations were very small. At the position where the ITO stress was 0, the theoretical and simulation values for the critical modulus of the film were both 20 GPa. The ITO was within the safe region under the five Young's moduli tested. As shown in Fig. 5f, the critical radii of the films were 2.2 mm and 1.0 mm when the Young's moduli for the films were 5 GPa and 10 GPa, respectively, and failure did not occur in the devices when the rest moduli were above the bending radius of 1.0 mm. The stress distribution curve under each modulus was not the same as that in Fig. 4f for the linear symmetric distribution when the stress was 0. Clearly, in order to make the distribution of the curve symmetrical, a larger difference in the Young's modulus is required, which also means that it is much more difficult to achieve a large shift in the position of the NA by changing the modulus compared with changing the thickness.

According to these results, it is feasible and effective to predict the location of a single NL based on the FEA method, especially for structures containing many membrane layers. Thus, in order to achieve the same protective effect, changing the thickness of the material is a better choice than modulating the Young's modulus, which is also known as NL thickness optimization design.

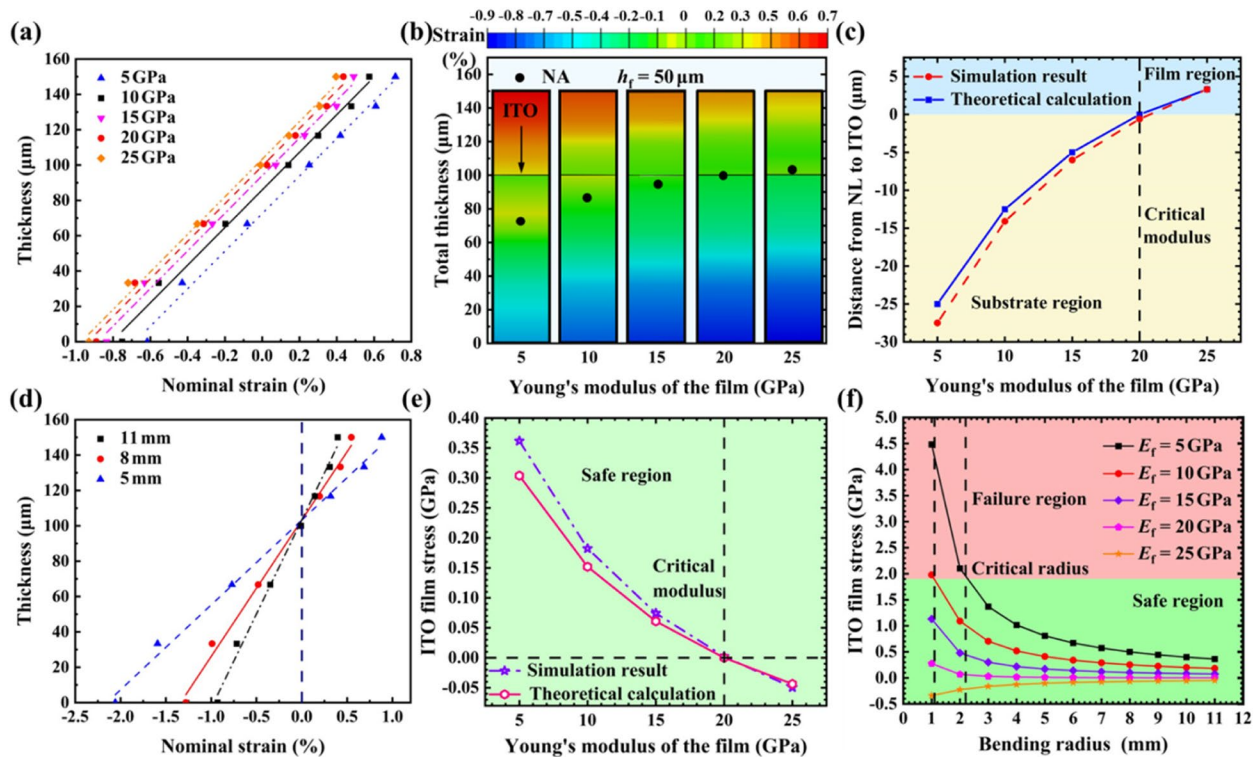


Fig. 5 Nominal strain distribution in structure with different values for: **a** Young's modulus of film, and **d** bending radius ($h_f = 50 \mu\text{m}$). Position of NL and stress on ITO in structure: **b** position of NA in

thickness direction, **c** distance from NL to ITO, **e** stress in ITO (11 mm bending radius), and **f** stress in ITO under different bending radii

4.2 Simulation of structures with multiple NLs

As mentioned above, multiple NLs can appear in a device by introducing OCA. In general, the strain is not uniformly distributed along the thickness direction in a structure with multiple NLs, so the theoretical calculation method is not suitable for determining the position of each single NL. Several previous studies have confirmed the existence of multiple NLs in a device based on experiments [56], but it is undoubtedly very difficult to determine the numbers and locations of NLs in devices with more complex structures. However, compared with mathematical calculations, the FEA method can extract the stress and strain situation for each layer in a device [57], and thus help us to more easily determine the locations and numbers of NLs. Therefore, we then replaced the ITO with OCA based on a single NL structure to connect the film and substrate, and simulated the locations and numbers of multiple NLs as well as the impact of OCA on the overall structure of the device by using the FEA method.

4.2.1 Influence of OCA thickness

In order to study the effect of the thickness of the OCA on the NL, models of the film, OCA, and substrate were first

established. The parameters for each material are shown in Table 5. The properties of the OCA can be defined by parameters describing hyperelasticity and viscoelasticity. The parameters for the OCA were first based on previously reported data [3]. All of the parameters for the film and substrate were kept unchanged, and only the thickness of the OCA was changed. The initial bending radius was 11 mm.

Only one NL appeared when the structures were bonded through an adhesive layer with an elastic modulus (5 GPa) that was much higher than that for the OCA. However, after the OCA was introduced, two NLs appeared in the device beside the OCA, and they were located in the film and substrate (see Fig. 6a). The decoupling effect of OCA [3] meant that the strain in the device was not completely distributed evenly along the thickness direction.

Table 5 Material properties and OCA thickness used in FEA

Labels (from top to bottom)	Material	$E(\text{GPa})$	ν	$h(\mu\text{m})$
Film	SiN_x film	5	0.29	100
OCA	OCA	/	0.50	25/50/75/100/125
Substrate	PI substrate	5	0.29	100

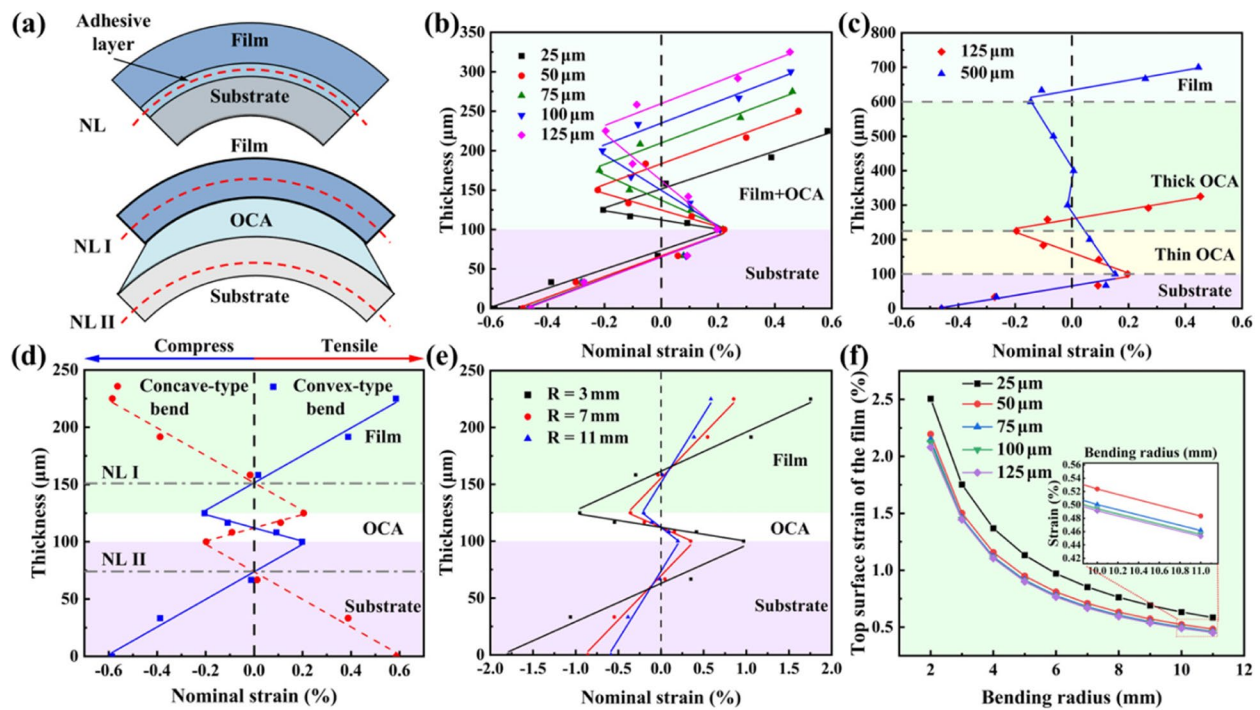


Fig. 6 **a** Position of NL in structure containing an adhesive layer (top) and OCA (bottom). **d** Strain distribution for different thicknesses in a device containing OCA with a thickness of 25 μm . Strain distribution in thickness direction when: **b** attached to OCA with different thick-

nesses, **c** attached to OCA with thicknesses of 125 μm and 500 μm , and **e** attached to OCA with a thickness of 25 μm at different bending radii. **f** Change in strain on top surface of the film with the bending radius

The strain in each film on the symmetric surface changed almost uniformly along the thickness direction, whereas it did not between the films (see Fig. 6d). For the convex-type bend, both tensile and compressive strains were found in the substrate, and thus there was an NL in the substrate. The maximum tensile and compressive strains were almost equal in the OCA, indicating that the NL for the OCA was in the middle thickness. Similarly, an NL appeared in the film. In addition, the stress conditions were completely opposite for the film and substrate, which means that OCA can divide connected films and substrates with the same material parameters into two identical and independent parts. Figure 6d also shows the strain distribution in the device under the concave-type bend, where the strain was also symmetrically distributed along a strain of 0 under the two bending modes.

As the thickness of the OCA increased, the maximum tensile strain on the top surface of the film decreased and the range decreased gradually, where the strain was almost equal with thicknesses of 100 μm and 125 μm (see Fig. 6b), before decreasing as the thickness of the OCA increased to 500 μm (see Fig. 6c), and the strain distribution when the thickness of the OCA was 125 μm was also selected. Clearly, the maximum tensile strain on the top surface of the film was almost the same under the two thicknesses.

As the bending radius increased, the maximum tensile and compressive strains decreased in both the film and substrate, and the NLs of the two film layers gradually approached the OCA (see Fig. 6e). As the bending radius approached infinity, the device approximated the unbending state, at which point the maximum tensile and compressive strains in both the film and substrate tended toward 0, thereby matching the strain distribution in an unbent device. As the bending radius decreased, the strain on the top surface of the film increased continuously at the five different thicknesses, and decreased as the thickness of the film increased under the same bending radius (see Fig. 6f), indicating that the bending radius does not change the influence of OCA thickness on the top surface strain of film.

4.2.2 Influence of OCA elastic modulus

In order to explore the effect of the elastic modulus of the OCA on the NL, a bending simulation was conducted with an initial bending radius of 11 mm for the device where the elastic modulus of the OCA was changed. The material parameters for the film, OCA, and substrate are shown in Table 5, and the thickness of the OCA was 25 μm . The two viscoelastic parameters for the OCA (g_i , τ_i) were kept unchanged and only the hyperelastic parameters (C_{10} , C_{20} , C_{30}) changed to reflect the changes in the elastic

modulus of the OCA. The simulation results were obtained using one, two, three, four, and five times the original values of the parameters. The simulation results are shown in Fig. 7a, where E represents the initial elastic modulus for the OCA ($E = 6C_{10}$).

As shown in Fig. 7a, as the elastic modulus of the OCA increased, the maximum tensile and compressive strains on the top surface of the film and the bottom surface of the substrate increased continuously, and the two NLs on the film and substrate were also constantly close to the OCA. According to these simulation results, increasing the elastic modulus of the OCA increased the maximum strain. OCAs with original moduli of 10 , 10^2 , 10^3 , 10^4 , and 10^5E were used, and the OCA with a modulus of 10^5E had an initial elastic modulus of ~ 4 GPa, which was close to that for the adhesive layer (5 GPa).

Figure 7b shows that when the modulus of the OCA was 10 times the original value, the strains on the bottom surface of the film and the top surface of the substrate were close to 0. As the OCA modulus continued to increase, tensile and compressive strains appeared in the film and substrate, respectively. The modulus increased at a rate almost an order of magnitude higher, but the strain distribution was almost the same from 10^2E OCA, although the structure containing an OCA with a higher modulus had greater strain at the same

thickness, and the strain distribution in the OCA with 10^5E was very close to that of the adhesive layer. In conclusion, when the elastic modulus of OCA was small, increasing the elastic modulus of the OCA increased the maximum strain. To a certain extent, when the elastic modulus was large, the strain began to exhibit a linear distribution along the thickness and the strain was almost equal at the same thickness, and the multiple NLs in the structure evolved into a single NL.

For OCA with 1–10 times the original modulus, when the modulus was larger, the strain on the surface of the film was larger at the same bending radius (see Fig. 7c). The strain for the film with 10^3 times the original modulus was much larger than those with 1–10 times. When the modulus was greater than 10^3E , the structure was approximately in a state with a single NL.

In the model of the original modulus, the shear strain was very large at close to 200%, which was much larger than the strain on the surface of the film (see Fig. 7d). As the modulus of the OCA continued to increase, the shear strain gradually decreased, but the decrease was not large. From 10^2E to 10^5E , the maximum shear strain decreased significantly, and decreased to 1% (10^5E). The difference in the strain on the surface of the film was not large but the difference in the shear strain was very large. Thus, the elastic modulus of the

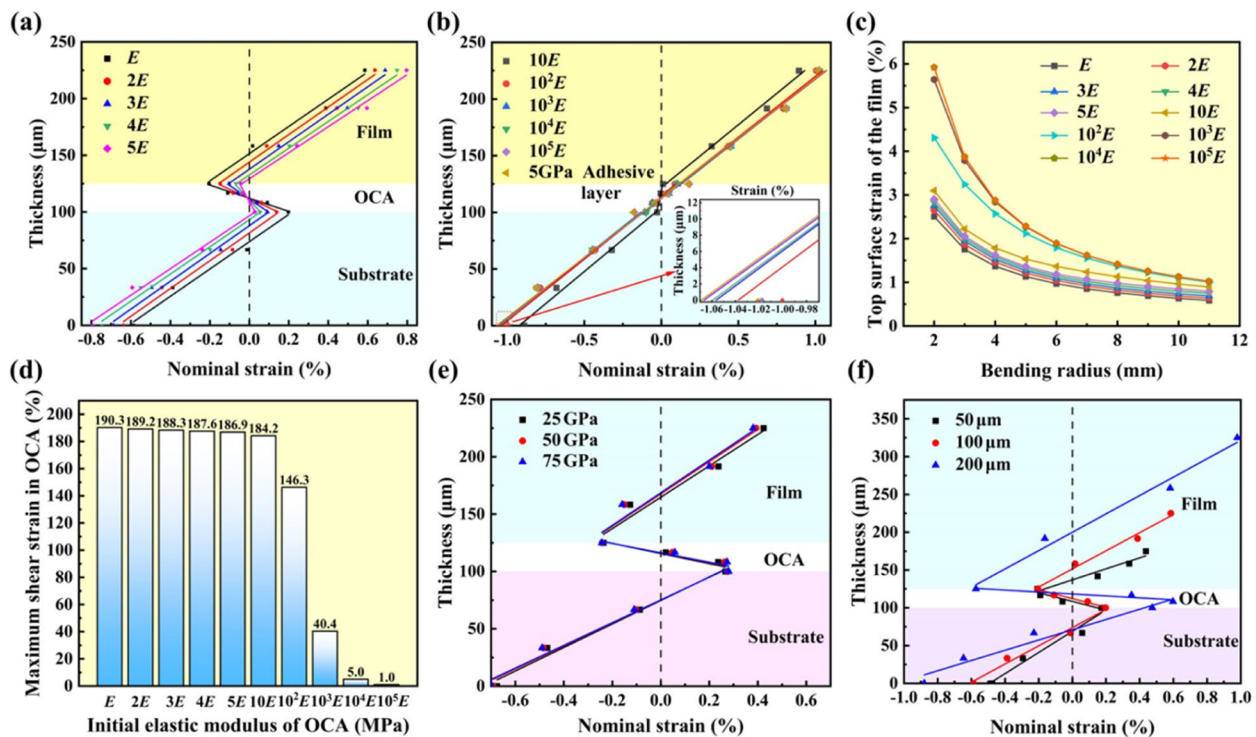


Fig. 7 Strain distribution in the thickness direction of the device: **a** using OCA with different moduli, and **b** using OCA with a higher modulus and adhesive layers. **c** Variations in strain on top surface of films containing OCA with different elastic moduli with bend-

ing radius. **d** Maximum shear strain in OCA varied with the elastic modulus (bending radius of 2 mm). Strain distribution in the thickness direction for devices with different film parameters: **e** Young's modulus, and **f** thickness

OCA had a great influence on its shear strain, and the shear strain was smaller when the modulus was higher. When the elastic modulus was very low, the structure was prone to delamination. When the modulus of the OCA was high, the maximum shear strain could be effectively reduced but the maximum strain in the device was also high, and the number of NLs in the structure changed; thus, the modulus of the OCA should be selected in a reasonable manner to meet the needs for safe device operation.

4.2.3 Effects of film parameters on multiple NLs

According to previous research, the position of a single NL in a device can be adjusted by changing the film parameters. Therefore, in order to further explore the effects of the film parameters on multiple NLs, simulations were conducted in this study. The material parameters in Table 5 were used and the thickness of OCA was selected as 25 μm . As the Young's modulus of the thin film increased, the strain distribution on the film, OCA, and substrate changed little (see Fig. 7e), thereby indicating that the modulus of the film did not affect the strain distribution, or influence the locations and numbers of NLs. The effect of the Young's modulus of the film on the NL was clearly different from that in a single NL structure.

As the film thickness increased, the maximum tensile strain of the top surface of the film increased and the position of the NL also increased gradually (see Fig. 7f), which are consistent with the effects of the film thickness in the single NL device. For a single film, changing the modulus did not change the position of the NL but it changed the thickness, so the position of the NL moved upward. Thus, the thin film was approximately independent due to the decoupling effect of the OCA.

4.3 Thickness optimization design for a single NL

The flexible electrode structure described in Sect. 3.1 was tested by using a dynamic bending test device. First, ITO films with the same thickness (~ 30 nm) were prepared on PI substrates with thicknesses of 25 μm , 50 μm , and 100 μm (see Fig. S3a). Each film was divided into three samples and tested three times. The corresponding resistance was recorded at different bending radii and the resistance was divided by the resistance before stretching for normalization. The bending forms were convex-type bend (ITO film tension) and concave-type bend (ITO film compression). The bending resistances of samples with substrate thicknesses of 25 μm , 50 μm , and 100 μm are shown in Figs. S3b–d. To simplify the presentation, the data in Figs. S3b–d are the averages based on the three tests. As the bending radius decreased, for the convex-type bend, the normalized resistance of the three samples increased slowly initially before

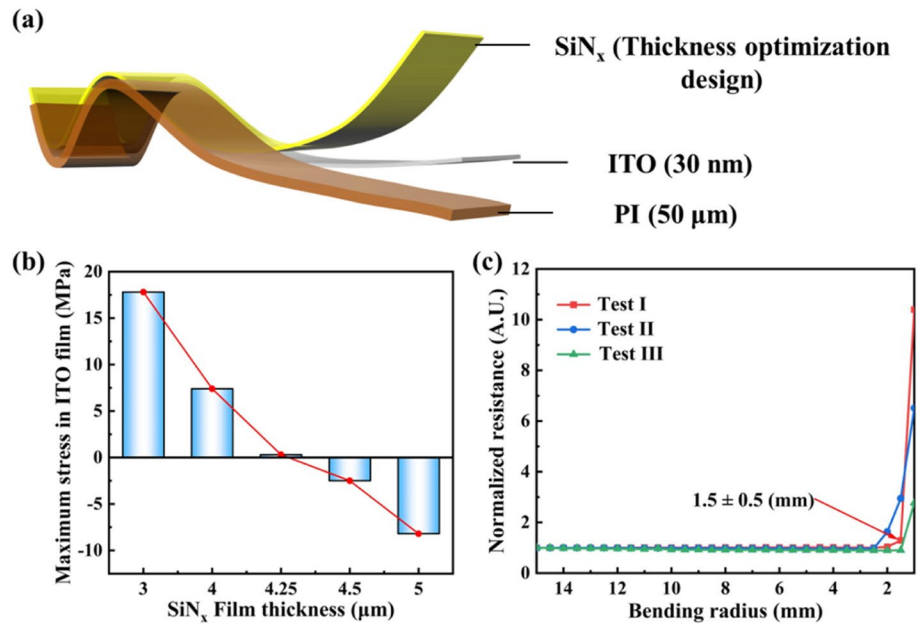
then increasing rapidly after reaching a certain radius. When the normalized resistance was greater than 1.1, i.e., the resistance was increased by 10% compared with the original resistance, crack propagation on the ITO became unstable due to tensile action [55]. The radius at this point was defined as the failure radius.

Under the convex-type bend, the failure radius was smallest for the sample with a substrate thickness of 25 μm at only 0.5 mm, and much larger (~ 4.2 mm) for the sample with a substrate thickness of 100 μm . Analysis of the three failure radii showed that the failure radius was larger when the substrate thickness was larger, which was as expected. According to the simulation results presented above, the NL gradually moved away from ITO as the substrate thickness increased, thereby increasing the ITO strain and making it fail at a larger radius. In addition, compared with the convex-type bend, the conductivity of the concave-type bend changed little for the samples with all three thicknesses, indicating that the ITO exhibited greater resistance to compression than stretching. The sample was bent to the failure radius in FEA simulations and the failure stress for the ITO was determined as ~ 1.9 GPa.

According to the simulation results presented above, the position of the NL could be changed by depositing a layer of SiN_x film on the ITO film. As shown in Fig. 8a, after optimizing the thickness of the film, the NL could be made to appear in the ITO, thereby reducing the failure radius for the ITO. When the PECVD method was used to deposit thicker films, the residual compressive stress was reduced, and thus the film cracked more easily. The NL also moved position as the modulus increased, so the SiN_x film with a larger modulus ($E = 249$ GPa) was identified as preferable, and a substrate thickness of 50 μm .

The parameters defined above were substituted into the simulation model and the thickness of the film was changed before conducting several simulations. The simulation results are shown in Fig. 8b. In the thickness optimization design process, the film thickness was varied at an interval of 1 μm (3 μm , 4 μm , and 5 μm). As the thickness increased, the maximum tensile stress gradually decreased in the ITO and became compressive stress. In order to make the ITO reach the ideal stress of 0, the thickness of the film was between 4–5 μm . The thickness was further optimized by division to 4.5 μm and then to 4.25 μm , and the maximum stress in the ITO was reduced to 0.3 MPa, which was very close to the ideal stress, and thus the optimal thickness of the film was determined as 4.25 μm . The composite film with the optimized thickness was assessed in three bending experiments and the experimental results are shown in Fig. 8c. The ITO failure radius was about 1.5 mm, which was about 50% smaller than that of 3.0 mm before optimization (see Fig. S3c), thereby indicating that the thickness optimization design process for a single NL was highly effective.

Fig. 8 **a** Structure of the film based on thickness optimization design. **b** Simulation results obtained by changing the maximum stress in the ITO film with the thickness of the SiN_x film. **c** Changes in normalized resistance of the structure with the bending radius after thickness optimization

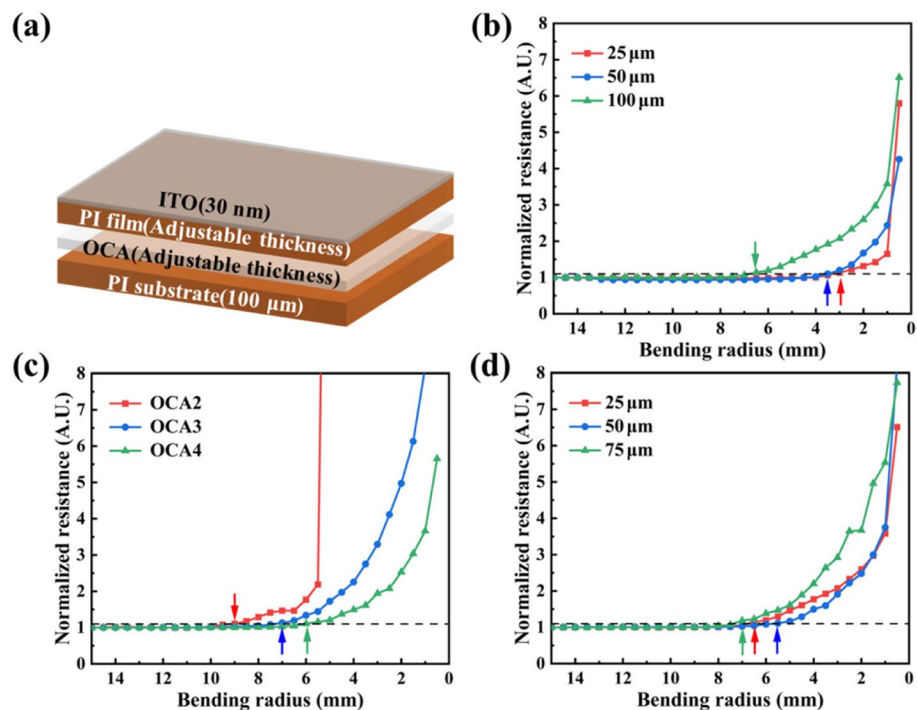


4.4 Sandwich electrode structure bending test

By bonding the flexible electrode and PI substrate using OCA, a sandwich electrode structure was obtained, as shown in Fig. 9a. The thickness of the bottom PI substrate was fixed at 100 μm , before adhering a layer of OCA and then a flexible electrode. The thickness of the PI film was adjusted. The changes in the resistance of the ITO were measured by using a dynamic bending device.

The thickness of the PI film was changed and OCA5 was used for the connection (see Fig. 9b). The failure radius was largest for the sample with a substrate thickness of 100 μm (~ 6.5 mm). The failure radius decreased further to ~ 3.5 mm for the substrate thickness of 50 μm and ~ 3 mm for the substrate thickness of 25 μm , which are consistent with the simulation results presented in the previous section, where the strain on the film top surface increased with the thickness (see Fig. 7f). Figure 9c shows the results obtained

Fig. 9 **a** Schematic diagram of composite electrode structure. Normalized resistance changes with the bending radius for: **b** PI films with different thicknesses, **c** different OCA types, and **d** different OCA thicknesses (the arrows indicate the failure radius for each ITO film)



after changing the type of OCA but keeping the thickness the same and using PI films with a thickness of 100 μm . OCA2 reached the failure standard when the bending radius was ~ 9 mm, and the failure radius for OCA3 (~ 7 mm) was very close to the failure radius for OCA4 (~ 6 mm). According to these simulation results, increasing the modulus of the OCA increased the maximum strain on the top film, and thus the failure radii for OCA3 and OCA4 were close but clearly smaller than that for OCA2. According to Fig. 9d, when OCA5 was used with different thicknesses, the failure radii were about 6.5 mm, 5.5 mm, and 7 mm at thicknesses of 25 μm , 50 μm , and 100 μm , respectively, which are very similar. The simulations presented in the previous section demonstrate that the thickness of the OCA had little influence on the maximum strain on the top layer film. Therefore, electrodes prepared using OCA with different thicknesses may fail at a similar bending radius.

4.5 Bending tests for electrode structures with multiple NLs

In general, many conductive layers are used in practical applications of flexible devices. Thus, in order to simulate a structure with multiple conductive layers, a double-layer flexible electrode was also designed containing two single-layer flexible electrodes. The preparation process is shown in Fig. 10. In the first step, a layer of single-sided PI tape was adhered to the upper surface of the single-layer ITO electrode. The thickness of the tape was 100 μm and the Young's modulus was ~ 1.10 GPa, which was close to the modulus of the PI substrate with a thickness of 100 μm (see Fig. 10a). In the second step, a 300- μm layer of double-sided PI tape ($E = 475$ MPa) or OCA1 was affixed to the surface of the single-sided PI tape (see Fig. 10b). In the third step, the structure shown in Fig. 10a was adhered to the double-sided tape or OCA1 surface to form a double-layer electrode structure. The electrodes were placed in the dynamic bending test device to perform the bending experiment. In order to more directly detect the changes in electrical conductivity in the two ITO layers, two bulbs were connected to the two ITO layers in the structure and the changes in performance were characterized by observing the light and dark changes in the bulbs, as shown in Figs. 10c and d.

As shown in Figs. 10c and d, the electrode adhered by double-sided PI tape was defined as structure I and the electrode adhered by OCA1 was defined as structure II. In the two structures, the positions of the upper ITO and lower ITO were designated as positions I and II, respectively, and connected with light bulbs I and II. According to Fig. 10e, when the bending radius reached 4.5 mm, the ITO at position I started to fail, and the resistance at position II changed little at 4.5 mm. Before bending, the brightness was relatively high for light bulbs I and II, but when the ITO at

position I began to fail, the resistance increased sharply and the brightness gradually declined for light bulb I, and it was extinguished when the bending radius was ~ 1 mm, whereas light bulb II remained lit but the brightness decreased significantly. As shown in Fig. 10f, the failure radii at positions I and II in structure II during the bending process were 2.5 mm and 3.0 mm, respectively, and thus significantly less than 4.5 mm. Due to the introduction of OCA1, multiple NLs were present in the structure, and the stress conditions for ITOs at the two positions were similar, so the failure radii were also similar. The brightness was relatively high for light bulbs I and II before bending, and the two bulbs were still bright when the bending radius was ~ 1 mm, although the brightness was lower, but the brightness was still higher than that for the same light bulb in structure I at the same bending radius.

The light and dark changes for the light bulbs at positions I and II were related to the strain in the ITO film. Bending simulations were conducted for structures I and II with a bending radius of 4.5 mm, and the strains were extracted at different thicknesses. The simulation results are shown in Fig. 11. Due to the large Young's modulus for the double-sided PI tape, an NL only appeared at the middle thickness for the double-sided PI tape in structure I. The strains in ITO1 and ITO2 at positions I and II were larger, reaching 5.6% and -5.5% , respectively, and ITO1 was subjected to large tensile strain and failed, thereby increasing the resistance and the light bulb was extinguished. ITO2 was subjected to a large compressive strain because its ability to resist compressive stress was greater than that for tensile stress, so the resistance increase was not as large as that for ITO1, and light bulb II was still lit after bending but the brightness was lower because the resistance increased. In structure II, due to the introduction of the low elastic modulus OCA1, two NLs were present in the structure, where the NLs appeared near ITO1 and ITO2. The maximum strain on the top layer of the single-sided PI tape also decreased from 10.2% to 3.1%. The strains were close to 0 for ITO1 and ITO2, but the increase in resistance was far less than that for ITO1 in structure I, and light bulbs I and II were lit after bending, and the brightness was higher. According to the simulation results, ITO1 and ITO2 in structure II were located in the NL and the strain was small, but there were certain deviations in the actual construction of the devices and the testing process, so the brightness also decreased to some extent for light bulbs I and II.

5 Conclusion

In this study, we investigated the failure behavior of fragile ITO films under bending in flexible electronic devices. In order to determine the failures caused by the mechanical

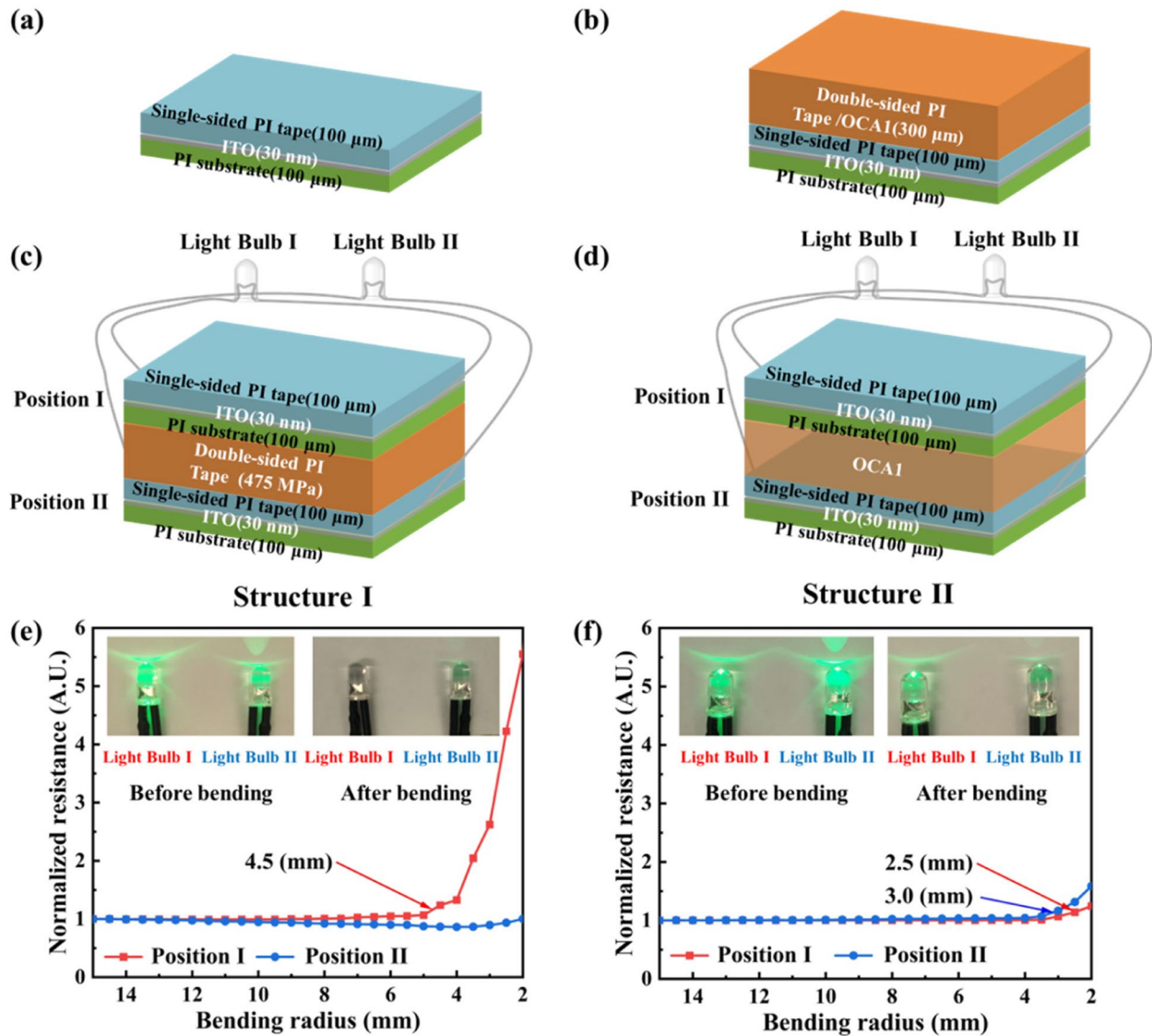


Fig. 10 **a** Attachment of single-sided PI tape. **b** Attachment of double-sided PI tape or OCA1. Double-layer flexible electrode structures I **c** and II **d**, as well as light bulbs I and II attached to the ITO. **e**, **f**

Normalized resistances of the structures at positions I and II varying with the bending radius

properties of the layered materials, analytical solutions were derived for the stress distributions in multilayer composite film structures under bending and nonlinear stress using the FEA method. The reliability of the proposed method was verified based on comparisons with experimental bending data. In addition, based on the mechanical model of OCA, the FEA method was used to analyze the numbers and locations of NLs in the multi-layer structure. Due to the decoupling effect of OCA, multiple NLs appeared in the structure. The effects of several key geometric and material design parameters on the NLs and device structures were also investigated. By measuring the change in the resistance of the ITO in the flexible electrode structure, the optimal thickness design was obtained for a

single NL and a method for using multiple NLs to maintain the ITO in a lower stress state.

The results showed that the NL gradually moved to the top surface of the film as the film thickness and Young's modulus increased. The introduction of OCA could make multiple NLs appear in the model, and reducing the elastic modulus of the OCA and the film thickness reduced the maximum strain in the film. When the elastic modulus was small for the OCA, the shear strain could be large in the model but adjusting the number of NLs helped to reduce the maximum shear strain.

The optimal design for the thickness of the NL obtained by using the simulation method could effectively reduce the failure of ITO electrodes. When the thickness of the

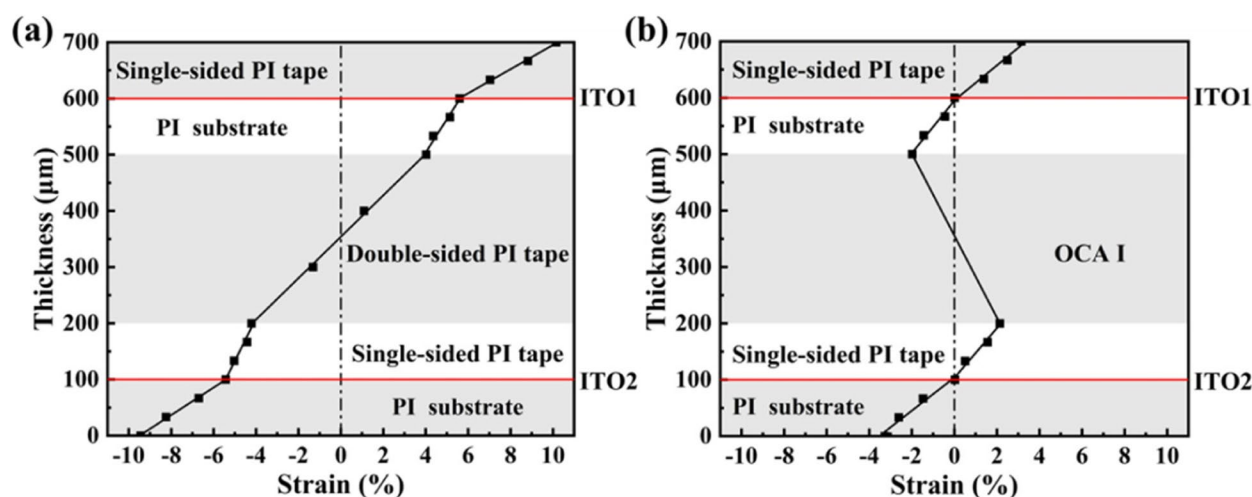


Fig. 11 Strain distributions for different structures along the thickness direction: **a** structure I, and **b** structure II

prepared SiN_x film was $4.25 \mu\text{m}$, the maximum stress on the ITO was reduced to 0.3 MPa and the failure bending radius decreased to 1.5 mm , which was 50% lower than that before optimization. The introduction of OCA into the double electrode structure could reduce the strain on the ITO from 5.6% to close to 0, and effectively protect the ITO. Thus, our practical verification and theoretical analyses indicate that the optimal thickness design for a single NL and the multiple NL strategy have broad application prospects, and they may provide ideal structural design schemes for flexible electronic devices.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s44275-024-00019-8>.

Acknowledgements Experimental supplies were provided by the Key Laboratory of Advanced Display and System Applications, Ministry of Education, Shanghai University, where experiments were also conducted. We also thank the Natural Science Foundation, China and Huawei Technologies Co., Ltd for their technical and financial support.

Authors' contributions Majiaqi Wu designed the study, conducted experiments, collated the data, used different methods for experimental verification and analysis, and finally wrote the original manuscript. Maoliang Jian, Jianhua Zhang, and Lianqiao Yang participated in data analysis, provided the corresponding revision suggestions, and approved the final draft.

Funding This research was financially supported by the Natural Science Foundation, China (Grant No. 52175512) and Huawei Technologies Co., Ltd.

Data availability Data and information about the materials used in this study are provided in the article.

Declarations

Ethics approval and consent to participate Not applicable.

Competing interests Jianhua Zhang is an Editorial Board Member. We declare that we have no financial and personal relationships with other people or organizations that could have inappropriately influenced our work, and there are no potential conflicts of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

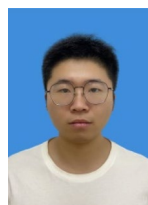
References

1. Jin DU, Kim TW, Koo HW, Stryakhilev D, Kim HS, Seo SJ et al (2010) 47.1: Invited paper: Highly robust flexible AMOLED display on plastic substrate with new structure. SID Symp Dig Tec Pap 41:703–705. <https://doi.org/10.1889/1.3500565>
2. Komatsu R, Nakazato R, Sasaki T, Suzuki A, Senda N, Kawata T et al (2014) 25.2: Repeatedly foldable book-type AMOLED display. SID Symp Dig Tec Pap 45:326–329. <https://doi.org/10.1002/j.2168-0159.2014.tb00088.x>
3. Jia Y, Liu Z, Wu D, Chen J, Meng H (2019) Mechanical simulation of foldable AMOLED panel with a module structure. Org Electron 65:185–192. <https://doi.org/10.1016/j.orgel.2018.11.026>
4. Mativenga M, Geng D, Kim B, Jang J (2015) Fully transparent and rollable electronics. ACS Appl Mater Interfaces 7:1578–1585. <https://doi.org/10.1021/am506937s>
5. Noda M, Kobayashi N, Katsuhara M, Yumoto A, Ushikura S, Yasuda R et al (2011) An OTFT-driven rollable OLED display. J Soc Inf Disp 19:316–322. <https://doi.org/10.1889/jsid19.4.316>

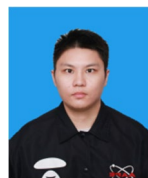
6. Gomes A, Priyadarshana LL, Visser A, Carrascal JP, Vertegaal R (2018) Magicscroll: a rollable display device with flexible screen real estate and gestural input. 20th International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI 2018) 6:1–11. <https://doi.org/10.1145/3229434.3229442>
7. Pang HQ, Rajan K, Silvernail J, Mandlik P, Ma RQ, Hack M et al (2011) Recent progress of flexible AMOLED displays. *Confer Adv Display Technol E-Papers Flexible Displays* 7956:79560J1–79610. <https://doi.org/10.1117/12.880144>
8. Jeong EG, Kwon JH, Kang KS, Jeong SY, Choi KC (2020) A review of highly reliable flexible encapsulation technologies towards rollable and foldable OLEDs. *J Inform Display* 21:19–32. <https://doi.org/10.1080/15980316.2019.1688694>
9. Kappaun S, Slugovc C, List EJ (2008) Phosphorescent organic light-emitting devices: working principle and iridium based emitter materials. *Int J Mol Sci* 9:1527–1547. <https://doi.org/10.3390/ijms9081527>
10. Suh S, Yi K, Choi C, Park D, Kim C (2012) Mobile LCD device with transparent infrared image sensor panel for touch and hover sensing. *IEEE Int Confer Consum Electron (ICCE)* 2012:213–214. <https://doi.org/10.1109/ICCE.2012.6161834>
11. Kim DS, Ko YJ, Park SM, Jung JH, Shestak S (2010) Stereoscopic display technologies for FHD 3D LCD TV. *Three-Dimensional Imaging, Visualization, and Display 2010 and Display Technologies and Applications for Defence, Security, and Avionics IV* 7690:769008–1–8. <https://doi.org/10.1117/12.850656>
12. Takasugi S, Shin HJ, Chang MK, Ko SM, Park HJ, Lee JP et al (2016) Advanced compensation technologies for large-sized UHD OLED TVs. *J Soc Inf Disp* 24:410–418. <https://doi.org/10.1002/jsid.442>
13. Han CW, Park JS, Choi HS, Kim TS, Shin YH, Shin HJ et al (2014) Advanced technologies for UHD curved OLED TV. *J Soc Inf Disp* 22:552–563. <https://doi.org/10.1002/jsid.287>
14. Chang YL, Lu ZH (2013) White organic light-emitting diodes for solid-state lighting. *J Disp Technol* 9:459–468. <https://doi.org/10.1109/JDT.2013.2248698>
15. Liu YF, Feng J, Bi YG, Yin D, Sun HB (2019) Recent developments in flexible organic light-emitting devices. *Adv Mater Technol* 4:1800371. <https://doi.org/10.1002/admt.201800371>
16. Guo KP, Tang Z, Chou XX, Pan SH, Wan CC, Xue T et al (2023) Printable organic light-emitting diodes for next-generation visible light communications: a review. *Adv Photonics Nexus* 2:044001–044014. <https://doi.org/10.1117/1.apn.2.4.044001>
17. Dai ZY, Lei M, Ding S, Zhou Q, Ji B, Wang MR et al (2024) Durable superhydrophobic surface in wearable sensors: From nature to application. *Explor* 4:20230046. <https://doi.org/10.1002/exp.20230046>
18. Wang XM, Qi LQ, Yang HH, Rao Y, Chen HP (2023) Stretchable synaptic transistors based on the field effect for flexible neuromorphic electronics. *Soft Sci* 3:1–20. <https://doi.org/10.20517/ss.2023.06>
19. Liu ZY, Wang DY, Gao H, Li MX, Zhou HX, Zhang C (2023) Metasurface-enabled augmented reality display: a review. *Adv Photonics* 5:034001–034029. <https://doi.org/10.1117/1.ap.5.3.034001>
20. Zhang YD, Miller DC, Bertrand JA, Jen SH, Yang R, Dunn ML et al (2008) 12.3: defect visualization of atomic layer deposition enabled polymer barriers using fluorescent tags. *SID Symp Dig Tech Pap* 39:143–146. <https://doi.org/10.1889/1.3069422>
21. Park EK, Kim S, Heo J, Kim HJ (2016) Electrical evaluation of crack generation in SiN_x and SiO_xN_y thin-film encapsulation layers for OLED displays. *Appl Surf Sci* 370:126–130. <https://doi.org/10.1016/j.apsusc.2016.02.142>
22. Chen Z, Cotterell B, Wang W (2002) The fracture of brittle thin films on compliant substrates in flexible displays. *Eng Fract Mech* 69:597–603. [https://doi.org/10.1016/s0013-7944\(01\)00104-7](https://doi.org/10.1016/s0013-7944(01)00104-7)
23. Park YT, Kim S, Ham SB, Cho SM (2020) Folding-stability criteria of thin-film encapsulation layers for foldable organic light-emitting diodes. *Thin Solid Films* 710:138277. <https://doi.org/10.1016/j.tsf.2020.138277>
24. Van Der Sluis O, Abdallah AA, Bouten PCP, Timmermans PHM, Den Toonder JMJ, De With G (2011) Effect of a hard coat layer on buckle delamination of thin ITO layers on a compliant elastoplastic substrate: An experimental–numerical approach. *Eng Fract Mech* 78:877–889. <https://doi.org/10.1016/j.engfracmech.2011.01.013>
25. Lee CC, Tsai CC, Chuang JC, Huang PC, Cheng SW, Liou YY (2016) Adhesion investigation of stacked coatings in organic light-emitting diode display architecture. *Surf Coat Technol* 303:226–231. <https://doi.org/10.1016/j.surfcoat.2016.03.041>
26. Auch MDJ, Soo OK, Ewald G, Soo-Jin C (2002) Ultrathin glass for flexible OLED application. *Thin Solid Films* 417:47–50. [https://doi.org/10.1016/s0040-6090\(02\)00647-8](https://doi.org/10.1016/s0040-6090(02)00647-8)
27. Schwamb P, Reusch T, Brabec CJ (2013) Flexible top-emitting OLEDs for lighting: Bending limits. *Confer Organic Light Emit Mater Devices XVII* 8829:88291E-1–88310. <https://doi.org/10.1117/12.2025011>
28. Cho TY, Lee WJ, Lee SJ, Lee JH, Ryu J, Cho SK et al (2018) Moisture barrier and bending properties of silicon nitride films prepared by roll-to-roll plasma enhanced chemical vapor deposition. *Thin Solid Films* 660:101–107. <https://doi.org/10.1016/j.tsf.2018.06.003>
29. Han YC, Jeong EG, Kim H, Kwon S, Im HG, Bae BS et al (2016) Reliable thin-film encapsulation of flexible OLEDs and enhancing their bending characteristics through mechanical analysis. *RSC Adv* 6:40835–40843. <https://doi.org/10.1039/c6ra06571f>
30. Choi GC, Kim DE, Park JW, Kim BS, Kwon YS (2009) Efficiency improvement of OLED by aquaregia and RCA treatment of ITO substrate. *Mol Cryst Liq Cryst* 504:35–43. <https://doi.org/10.1080/15421400902939041>
31. Katam NK, Singh C, Rawat M, Anand RS (2013) Effect of chemical treatments on ITO and OLED device. *Adv Mater Res* 849:387–390. <https://doi.org/10.4028/www.scientific.net/AMR.849.387>
32. Park SK, Han JI, Moon DG, Kim WK (2003) Mechanical stability of externally deformed indium-tin-oxide films on polymer substrates. *Jpn J Appl Phys* 42:623–629. <https://doi.org/10.1143/jjap.42.623>
33. Chiang CJ, Winscom C, Bull S, Monkman A (2009) Mechanical modeling of flexible OLED devices. *Org Electron* 10:1268–1274. <https://doi.org/10.1016/j.orgel.2009.07.003>
34. Lee S, Kwon JY, Yoon D, Cho H, You J, Kang YT et al (2012) Bendability optimization of flexible optical nanoelectronics via neutral axis engineering. *Nanoscale Res Lett* 7:7. <https://doi.org/10.1186/1556-276x-7-256>
35. Lee CC, Liou YY (2018) Dependent analyses of multilayered material/geometrical characteristics on the mechanical reliability of flexible display devices. *IEEE Trans Device Mater Reliab* 18:639–642. <https://doi.org/10.1109/tdmr.2018.2878485>
36. Niu YF, Liu SF, Chiou JY, Huang CY, Chiu YW, Lai MH et al (2016) Improving the flexibility of AMOLED display through modulating thickness of layer stack structure. *J Soc Inf Disp* 24:293–298. <https://doi.org/10.1002/jsid.443>
37. Lee CC, Shih YS, Wu CS, Tsai CH, Yeh ST, Peng YH et al (2012) Development of robust flexible OLED encapsulations using simulated estimations and experimental validations. *J Phys D: Appl Phys* 45:275102. <https://doi.org/10.1088/0022-3727/45/27/275102>
38. Campbell CJ, Clapper J, Behling RE, Erdogan B, Beagi HZ, Abrahamson JT et al (2017) P-198: Optically clear adhesives enabling

- foldable and flexible OLED displays. SID Symp Dig Tec Pap 48:2009–2011. <https://doi.org/10.1002/sdtp.12071>
39. Lin L, Li Y, Hu K, Yuan B, Huang G, Gao X et al (2018) 68–1: Invited paper: Reliability and failure mode analysis of foldable AMOLED display module. SID Symp Dig Tec Pap 49:899–901. <https://doi.org/10.1002/sdtp.12250>
 40. Nam J, Lee S, Han M, Lee H (2021) Improved stack structure of rollable display to prevent delamination and permanent deformation. Int J Precis Eng Manuf 22:671–678. <https://doi.org/10.1007/s12541-021-00481-6>
 41. Wang WQ, Jia YZ, Li HQ, Jiang MY, Zhang Z (2021) Mechanical simulation of foldable organic light-emitting diode display supporting layer. J Soc Inf Disp 29:723–730. <https://doi.org/10.1002/jsid.1028>
 42. Shi S, Li Z, Dong L, Du S, Tsai PM, Wang W et al (2019) 18.3: Invited paper: Research on 7.56-inch foldable AMOLED and relevant foldable technologies. SID Symp Dig Tec Pap 50:184–186. <https://doi.org/10.1002/sdtp.13433>
 43. Kim W, Lee I, Kim DY, Yu YY, Jung HY, Kwon S et al (2017) Controlled multiple neutral planes by low elastic modulus adhesive for flexible organic photovoltaics. Nanotechnol 28:9. <https://doi.org/10.1088/1361-6528/aa6a44>
 44. Walls R, Viljoen C, De Clercq H (2019) A nonlinear, beam finite element with variable, eccentric neutral axis. Eng Struct 187:341–351. <https://doi.org/10.1016/j.engstruct.2019.02.056>
 45. Suo Z, Ma EY, Gleskova H, Wagner S (1999) Mechanics of rollable and foldable film-on-foil electronics. Appl Phys Lett 74:1177–1179. <https://doi.org/10.1063/1.123478>
 46. Kim YG, Park BM, Choi JK, Jang DH, Han KY (2021) Optimization of the rework of bended OLED displays by surface-energy control. Coatings 11:1523. <https://doi.org/10.3390/coatings11121523>
 47. Wang M, Chen DB, Feng W, Zhong AY (2015) Synthesis and characterization of optically clear pressure-sensitive adhesive. Mater Trans 56:895–898. <https://doi.org/10.2320/matertrans.M2014425>
 48. Kossa A, Valentine MT, Mcmeeking RM (2023) Analysis of the compressible, isotropic, Neo-Hookean hyperelastic model. Meccanica 58:217–232. <https://doi.org/10.1007/s11012-022-01633-2>
 49. Peng X, Han L, Li L (2021) A consistently compressible Mooney-Rivlin model for the vulcanized rubber based on the Penn's experimental data. Polym Eng Sci 61:2287–2294. <https://doi.org/10.1002/pen.25757>
 50. Melly SK, Liu L, Liu Y, Leng J (2022) Modified Yeoh model with improved equibiaxial loading predictions. Acta Mech 233:437–453. <https://doi.org/10.1007/s00707-021-03105-2>
 51. Yu LD, Jin TF, Yin ZN, Xiao H (2015) Bounded multi-axial elastic potentials with strain-stiffening effects for rubbery materials based on uniaxial data. Chin J Solid Mech 36:164–170. <https://doi.org/10.19636/j.cnki.cjasm42-1250/o3.2015.02.009>
 52. Wu D, Liao D, Shi J, Jia Y (2021) Structural optimization method of foldable active-matrix organic light-emitting diode panel based on mechanical theory. J Soc Inf Disp 29:923–929. <https://doi.org/10.1002/jsid.1015>
 53. Slanik ML, Nemes JA, Potvin MJ, Piedboeuf JC (2000) Time domain finite element simulations of damped multilayered beams using a prony series representation. Mech Time-Depend Mater 4:211–230. <https://doi.org/10.1023/a:1009826923983>
 54. Leterrier Y, Mottet A, Bouquet N, Gilliéron D, Dumont P, Pinyol A et al (2010) Mechanical integrity of thin inorganic coatings on polymer substrates under quasi-static, thermal and fatigue loadings. Thin Solid Films 519:1729–1737. <https://doi.org/10.1016/j.tsf.2010.06.003>
 55. Alzoubi K, Hamasha MM, Lu S, Sammakia B (2011) Bending fatigue study of sputtered ITO on flexible substrate. J Disp Technol 7:593–600. <https://doi.org/10.1109/JDT.2011.2151830>
 56. Nishimura M, Takebayashi K, Hishinuma M, Yamaguchi H, Murayama A (2019) A 5.5-inch full HD foldable AMOLED display based on neutral-plane splitting concept. J Soc Inf Disp 27:480–486. <https://doi.org/10.1002/jsid.796>
 57. Feng Q, Su K, Du G, Wang Z, Lyu G (2021) P-55: Analysis of bonding optically clear adhesive in dual-cell LCD. SID Symp Dig Tec Pap 52:1275–1278. <https://doi.org/10.1002/sdtp.14933>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Majiaqi Wu (1996–), from Ningbo, Zhejiang Province. Doctoral candidate. He received his Master's degree from Shanghai University, China in 2023. His main research interest is inorganic thin film materials.



Maoliang Jian (1997–), from Xinyang, Henan Province. Doctoral candidate. He received his Bachelor's degree from Shanghai Maritime University, China in 2019. His main research interests are thermal management and the reliability of power devices.



Jianhua Zhang (1972–), from Hefeng, Hubei Province. Professor, Doctoral supervisor. She received her Ph.D. from Shanghai University, China in 1999. Her research interests include new AMOLED display, flexible display, and thin film encapsulation, thin film transistor display, and sensing.



Lianqiao Yang (1979–), from Baoding, Hebei Province. Professor, Doctoral supervisor. She received her Ph.D. from Myongji University, Korea in 2009. Her research interests include thermal management and reliability of power semiconductor devices, advanced optoelectronic materials, and device fabrication.